Anup Sedhain

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Experience

Sr. Fullstack AI Engineer, Airvet Inc. – Los Angeles, CA Jan 2026 – Present

- Built and shipped Atlas, a clinical AI assistant that puts peer-reviewed veterinary literature – 2,800+ clinical sources spanning 20+ publishers – at the point of care, replacing hours of internet research with cited, evidence-backed answers vets can act on.
- Made every answer clinically trustworthy with a strict citation validator that traces each sentence back to its source – catching unsupported claims in nearly 1 in 4 draft answers before they reach clinicians – backed by schema-validated structured outputs and automated groundedness scoring.
- Improved retrieval F1 by 23% with a dual-query reranking pipeline (Cohere Rerank + Reciprocal Rank Fusion) over hybrid vector+keyword search across 4 knowledge bases.
- Turned evals into product decisions with golden-set experiments and end-to-end tracing (Langfuse), shipping an answer-comparison (A/B choice) pattern to improve response quality.

Software Engineer, Airvet Inc. – Los Angeles, CA May 2025 – Dec 2025

- Cut feature-to-release cycles from 4 to 2 weeks and manual QA time by 80% by automating the development pipeline across JIRA, Figma, and GitHub with MCP-based AI agents; maintained a 99.9% deployment success rate.
- Ensured case updates reliably reach vet-techs during live consultations as owner of Airvet's Care Portal (Next.js, TypeScript, Go), redesigning the SSE event architecture with state reconciliation and cross-channel deduplication.
- Led development of Airvet's parent-facing web application (Next.js, Go, Twilio, AWS) and the early RAG research that seeded the company's veterinary AI roadmap.

Graduate Research Assistant, University of Florida – Gainesville, FL Jan 2025 – Jun 2025

- Achieved 92% accuracy in AI-content detection across 2M+ BlueSky events by building a multistage NLP pipeline (BERT, K-means, TF-IDF) on the HiPerGator HPC cluster, with temporal network analysis to model attitude evolution.

Software Engineer, Airvet Inc. – Los Angeles, CA May 2021 – Jul 2024

- Enabled vet-techs to handle 100+ concurrent cases and cut response times by 75% by building a real-time, event-driven Care Portal (React, Golang, SSE, Redis pub/sub) with sub-200ms latency and 99.9% uptime.
- Modernized airvet.com by replacing a legacy system with a Next.js application backed by headless WordPress, letting marketing publish content without engineering support.
- Drove near-100% B2B client retention by building an analytics ecosystem (Firebase, Amplitude, LogRocket) and admin tools that turned usage data into product iterations.

Software Engineer, Freelance, Polydelic AS – Oslo, Norway Aug 2020 – Jun 2021

- Helped clients secure millions in funding by delivering 6+ SaaS platforms (React, Node.js, GraphQL, AWS) serving 500K+ users at 99.9% uptime.
- Scaled a video streaming platform to 100+ concurrent users with a mediasoup-based SFU, load balancing, and Redis caching.
- Launched Rubato, an artist-booking marketplace processing €100K+ in monthly transactions (ElasticSearch, GraphQL, automated contract processing).

Software Engineer, Leapfrog Technology Inc. – Kathmandu, Nepal Jul 2019 – Dec 2020

- Accelerated delivery timelines by 40% across 50+ projects by building an enterprise project-management platform with automated workflows (Firebase, LitElement).
- Improved protein-workflow processing efficiency by 25% for 200+ monthly batches on a bio-informatics platform (Micronaut, MySQL).

Education

University of Florida, Master of Science in Computer Science, GPA 3.92 — Aug 2024 – Dec 2025

Kathmandu University, Bachelor of Computer Engineering, GPA 3.63 — Sep 2015 – Jul 2019

Technical Skills

Applied AI: LLM Agents, RAG, Reciprocal Rank Fusion, Hybrid Search & Reranking, Prompt Engineering, Tool Use & MCP, Structured Outputs, Guardrails, LLM Evals & Observability (Langfuse, Braintrust), Vector Databases, OpenAI API, AWS Bedrock, Hugging Face, Ollama

Programming: Python, TypeScript, JavaScript, Go, C++, Node.js, Java, Rust, OpenMP, MPI

Frontend: React, Next.js, Redux, Tailwind CSS, WebRTC, WASM, Real-time Systems

Backend & Data: PostgreSQL, MySQL, MongoDB, Redis Pub/Sub, Event-Driven Architectures, Server-Sent Events, Firebase, ElasticSearch, RESTful APIs, GraphQL

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Kafka, Observability & Monitoring